

INFO9015 (Logic for Computer Science): Sudokus Solver

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1 Sudoku Solver

Here is all the constraints to solve a Sudoku.

1.1 Sudoku of size 9x9 & 4x4 :

- Set all the possible values for a case (1 to N) :

$$\bigwedge_{i=1}^N \bigwedge_{j=1}^N \left(\bigvee_{k=1}^N \text{newlit}(i, j, k) \right)$$

- Must be different values in a row of the matrix (Sudoku) :

$$\bigwedge_{i=1}^N \left(\bigwedge_{kRow=1}^N \bigwedge_{j=1}^N \bigvee_{jAlt=j+1}^N (\neg \text{newlit}(i, j, kRow)) (\neg \text{newlit}(i, jAlt, kRow)) \right)$$

- Must be different values in a column of the matrix (Sudoku) :

$$\bigwedge_{i=j}^N \left(\bigwedge_{k=1}^N \bigwedge_{i=1}^N \bigvee_{iAlt=j+1}^N (\neg \text{newlit}(i, j, k)) (\neg \text{newlit}(iAlt, j, k)) \right)$$

- Must be different values in all sub-matrix :
Combine row and column constraints to each sub matrix of the sudoku
- Make sure that the specific constraints are included

1.2 Sudoku of size 16x16 & 25x25

- Same constraints that the Sudoku of size ≤ 9
- The new literals written in the .cnf file must have all the same length for i, j, and k to avoid confusion.

2 Sudoku has More Than One Solution

To verify if a Sudoku puzzle has more than one solution, the process involves excluding the current solution from the Sudoku in CNF (Conjunctive Normal Form). Subsequently, additional constraints are created, and a new instance of the Sudoku is solved. If this proves to be impossible, the Sudoku possesses a unique solution. Conversely, if a solution is found, it indicates that the Sudoku has more than one possible solution.

$$\bigwedge_{i=1}^N \bigwedge_{j=1}^N (\text{sudoku}[i-1][j-1] > 0 \rightarrow \neg \text{newlit}(i, j, \text{sudoku}[i-1][j-1]))$$

3 Creation of Sudoku

3.1 Normal creation

The process is straightforward. Initially, I select one of the Sudokus from the folder¹, and then, I iterate through the Sudoku, occasionally removing a single cell. The program ensures

1. random one

that deleting each cell maintains the uniqueness of the Sudoku by solving the sudoku in each iteration.

3.2 Sudokus with only `<size>` minus 1 numbers

The operation is identical to normal creation. However, the distinction lies in the program's initial removal of all cells where the value is equal to N before applying the standard creation algorithm.

4 A Personal Touch (Probabilistic Element)

4.1 Creation Difficulty

- I've incorporated a probabilistic element into the program. Users can opt² for a difficulty level, choosing from *"easy"*, *"medium"*, or *"hard"* using the following syntax :

```
python sudokub.py -c <size> <difficulty>
```

or

```
python sudokub.py -cm <size> <difficulty>
```

- If you increase the difficulty, the number of empty cells will increase, making the Sudoku more challenging to solve.
- Difficulty Levels :
 - Easy removes a maximum of 50% of the cells.
 - Medium removes a maximum of 75% of the cells.
 - Hard removes a maximum of 90% of the cells.

2. By default set to medium level